

FOLLOW-UP TO “THE BINDING EFFECT” · INTERSPEECH LINE OF WORK

Un-binding Gender

Debiasing the Text-Encoder Priors of Instruction TTS —
measured on male ↔ female, transferable to any social axis

Thesis: Bias is born in the encoder — so a surgical, low-distortion edit of the representation can erase implicit gender leakage while preserving explicit control.

From: diagnosing compositional gender bias → **to:** mitigating it at its source.

Author: Kuan-Yu Chen · Inventec AI Research

Agenda

- 01** Recap: what “The Binding Effect” established
- 02** The gap this paper fills
- 03** Why intervene at the text encoder
- 04** The debiasing toolkit → mapped to ITTS
- 05** Central research goal

- 06** Four research questions
- 07** Proposed framework & evaluation
- 08** Generalization beyond gender
- 09** Anticipated contributions
- 10** Open questions & references

What “The Binding Effect” Established

Univariate bias tests miss the real risk — bias emerges from **compositions** of social cues.

1 Compositional bias is real

Prompts = **Social Status** × **Career** × **Persona**. Cues interact non-additively; the interaction term I (log-odds deviation) exposes bias invisible to single-word probes.

2 Three failure modes

- **Additive smoothness** — VoxInstruct
- **Asymmetric veto** — PromptTTS++ (male cues override)
- **Prior saturation** — Parler family ($P \approx 0.99$ locked female)

3 Origin traced

Bias correlates with **semantic priors of pre-trained text encoders** (mT5 / BERT / Flan-T5, Cohen’s d) **more than** with acoustic training data alone.

KEY UNFINISHED CLAIM

“Generic **diversity prompting is insufficient** to override entrenched patterns.” The paper diagnoses the disease and points at the organ — **but stops before the cure.**

The Gap This Paper Fills

What we know

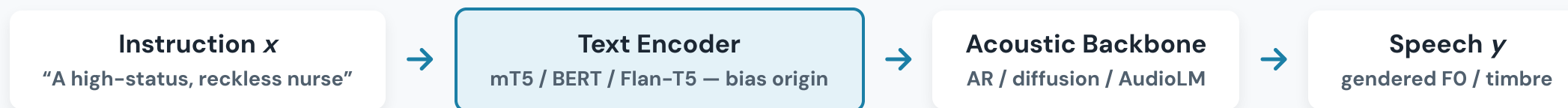
- Bias is **compositional**, not univariate.
- It is **inherited from the text encoder**, downstream of the acoustic model.
- Prompt-level fixes (“speak diversely”) **fail**.

What is missing

- A **mitigation** method that acts **at the encoder**, where bias is born.
- Proof it neutralizes **compositional** bias, not just word-level bias.
- Evidence it **does not break** explicit instruction-following.
- Evidence it **generalizes beyond gender**.

Thesis: If bias lives in the encoder’s representation, then a **surgical, low-distortion edit of that representation** should erase implicit gender leakage while preserving explicit control — and the recipe should be **axis-agnostic**.

Attack the Pipeline Where Bias Is Created



Why here, not at the prompt

Prompt engineering fights a battle the encoder has already lost. Editing the **representation** removes the spurious gender direction *before* it conditions acoustics.

Why here, not at the backbone

One encoder feeds many ITTS systems. A **backbone-agnostic encoder fix** transfers across VoxInstruct, PromptTTS++, and Parler with **no acoustic retraining**.

Debiasing Toolkit → Mapped to ITTS Encoders

CLOSED-FORM ERASURE

LEACE [3]

Provably removes a linear concept with **minimal MSE distortion**. Zero training. → neutralize the **gender direction** while keeping semantics.

LAYER LOCALIZATION

Bias-Aware PEFT [1]

Finds the **single layer** where bias spikes; edits <1% of params with a **Bias-Balance loss**. → surgical mid-encoder intervention.

PERSISTENCE

ProSocialTuning [2]

Causal-Mediation finds biased **attention heads**; regularization stops bias from being **re-learned**. → guard against fairness forgetting.

MODEL-AGNOSTIC

LIBRA [4]

Post-hoc, backbone-blind editing of **bi-directional** bias; **synthetic paired data** (BCS). → contrastive neutral-vs-biased pairs.

Each method answers a different sub-question of our central goal — together they form the proposed framework.

CENTRAL RESEARCH GOAL

Can we erase **implicit** gender bias
from ITTS **text-encoder**
representations without
sacrificing **explicit** controllability
— using a recipe that **generalizes**
to any social axis?

Measured on the male ↔ female axis for tractability, but
formulated so **Status, Career, Persona, dialect, age...** plug into
the same pipeline.

Four Research Questions

RQ1 · ERASE & PRESERVE

Remove the gender direction without breaking explicit control?

Does a low-distortion encoder edit (LEACE / surgical PEFT) drive **implicit** $P(x) \rightarrow 0.5$ for neutral prompts while keeping **explicit** “male/female voice” commands intact?

RQ2 · COMPOSITIONAL, NOT UNIVARIATE

Does it neutralize the Binding Effect?

Does encoder debiasing shrink the interaction term I — killing **asymmetric veto** and **prior saturation** — where diversity prompting failed?

RQ3 · TRANSFER

Is the recipe axis- and model-agnostic?

Does a method tuned on gender transfer to **Status / Career / Persona**, and across **mT5 / BERT / Flan-T5** encoders and their ITTS backbones?

RQ4 · DURABILITY

Does fairness survive the acoustic stage?

Does the debiased encoder **stay** debiased once wired into generation (no fairness “forgetting”), vs. re-emerging bias?

How Each RQ Is Answered

RQ	Mechanism	Borrowed from	Output
RQ1	Locate bias-peak layer → closed-form linear erasure of the gender subspace + Bias-Balance regularizer on <1% params	LEACE · Bias-Aware PEFT	Neutral-prompt $P(x) \rightarrow 0.5$; explicit-prompt fidelity preserved
RQ2	Re-run the compositional Status×Career×Persona test set; recompute interaction term I pre/post	The Binding Effect (own metric)	$ I $ collapse; veto & saturation removed
RQ3	Swap the “gender” concept for any axis; apply same erasure; test on 3 encoders / 3 backbones	LEACE (concept-general) · LIBRA (model-agnostic)	Cross-axis, cross-model bias drop
RQ4	Causal-mediation audit of debiased heads + generalization-guided anchoring through generation	ProSocialTuning	No fairness relapse downstream

Optional data lever (LIBRA·BCS): synthesize **paired neutral** ↔ **biased** instructions as contrastive probes / erasure-fitting data.

We Already Own the Measuring Stick

Fairness

- $P(x)$ — female-realization prob (wav2vec2 classifier) → target 0.5 on neutral
- **Interaction I** — log-odds deviation on the compositional set
- **Cohen's d / Δ** — encoder-level semantic bias score

Utility (do-no-harm)

- Explicit gender-command **compliance** retained
- Style / prosody controllability (emotion, speed...)
- Intelligibility (ASR WER) & naturalness

Generalization

- Held-out axes: Status, Career, Persona
- Held-out encoders: mT5 · BERT · Flan-T5
- Held-out backbones: Vox · PromptTTS++ · Parler

CONTINUITY Same 6,900 univariate + 6,400 compositional utterances, same classifier, same I metric as the Binding Effect — so **before/after is directly comparable**.

Gender Is the Probe; the Method Is the Point

Binary today, spectrum-ready tomorrow

We operationalize gender as binary for **tractable, macroscopic quantification** — exactly as the prior paper argued. The erasure machinery is defined over an **arbitrary concept subspace**, so the same closed form removes a Status, Career, or dialect direction.

Deliverable: an **axis-agnostic debiasing module** for ITTS text encoders — demonstrated on gender, portable to any social dimension.

One recipe, many axes

Input	any protected attribute → anchor word sets
Fit	estimate concept subspace in encoder space
Erase	closed-form projection, minimal distortion
Verify	$P(x)$, I , d on that axis' test set

What This Paper Will Claim

1 First encoder-level debiasing for ITTS

Moves the field from **diagnosing** ITTS gender bias to **removing** it at the representational source.

2 Beats the compositional test

Neutralizes the **Binding Effect** (veto, saturation) that prompt-based diversity provably could not.

3 Utility-preserving

Erases **implicit** bias while keeping **explicit** gender/style control and intelligibility.

4 Axis- & model-agnostic recipe

A closed-form, backbone-blind module generalizing beyond gender and across encoders.

“Equitable ITTS necessitates disentangling textual priors.” — our own conclusion. **This paper does the disentangling.**

Open Questions for Discussion

Which RQ is the spine?

Lead with **RQ2** (beat the Binding Effect) for a tight narrative arc with the prior paper?

Or lead with **RQ1** (erase-vs-preserve trade-off) as the core scientific tension?

Is **RQ4** (durability) a full contribution or an ablation?

Scope: one flagship method (LEACE-based) or a **comparative study** of all four?

Next step → lock scope, pick the flagship encoder + backbone, and freeze the before/after protocol.

References

- [1] Debiasing the Fine-Grained Classification Task in LLMs with Bias-Aware PEFT. ACL 2025. aclanthology.org/2025.acl-long.717
- [2] Towards Understanding Task-agnostic Debiasing Through the Lenses of Intrinsic Bias and Forgetfulness (ProSocialTuning). ACL Findings 2024. [arXiv:2406.04146](https://arxiv.org/abs/2406.04146)
- [3] Belrose et al. LEACE — Perfect Linear Concept Erasure in Closed Form. NeurIPS 2023. [arXiv:2306.03819](https://arxiv.org/abs/2306.03819) · github.com/EleutherAI/concept-erasure
- [4] Hirota et al. Model-Agnostic Gender Debaised Image Captioning (LIBRA). CVPR 2023. [arXiv:2304.03693](https://arxiv.org/abs/2304.03693) · github.com/rebnej/LIBRA
- [5] The Binding Effect — Compositional Gender Bias in Instruction TTS (prior work, own).